Ain Shams University

Faculty of Engineering

Computer and Systems Engineering Department

CSE356: Internet of Things

IoT Project Report

Smart Home Intrusion Detection System

Part 1: Complete Design of the IoT System

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Step 1: Purpose & Requirements

1.1 Purpose of the System

The **Smart Home Intrusion Detection System (SHIDS)** is an IoT-based security solution designed to monitor residential premises in real time and detect unauthorized entry or suspicious activity. The system leverages a network of sensors, cameras, and actuators connected through a central IoT gateway to provide homeowners with immediate alerts, remote monitoring capabilities, and automated responses to potential security threats — all accessible through a dedicated mobile and web application.

The primary goal of SHIDS is to enhance residential security by providing a cost-effective, scalable, and intelligent intrusion detection solution that bridges the gap between traditional alarm systems and modern smart home ecosystems. By combining real-time sensing, cloud connectivity, and intelligent alerting, the system empowers homeowners to monitor and protect their property from anywhere at any time.

1.2 Functional Requirements

Functional requirements define the specific behaviors and operations the system must perform. Table 1 lists the functional requirements of SHIDS.

Table 1: Functional Requirements of SHIDS

ID	Requirement
FR1	The system shall continuously monitor all entry points (doors, windows) using motion and contact sensors.
FR2	The system shall capture video and images upon detecting motion or unauthorized entry.
FR3	The system shall send real-time push notifications and alerts to the homeowner's mobile device upon intrusion detection.
FR4	The system shall trigger an audible alarm (siren) when an intrusion is confirmed.
FR5	The system shall allow the homeowner to arm and disarm the system remotely via a mobile application.
FR6	The system shall log all security events — including timestamps, sensor triggers, and snapshots — in a cloud database.
FR7	The system shall support live video streaming accessible to the homeowner remotely.
FR8	The system shall allow the homeowner to configure customizable alert zones and sensor sensitivity levels.
FR9	The system shall integrate with smart door locks to enable remote locking and unlocking.
FR10	The system shall notify designated emergency contacts or authorities in the event of a confirmed intrusion.

1.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the system. Table 2 lists the non-functional requirements of SHIDS.

Table 2: Non-Functional Requirements of SHIDS

ID	Category	Requirement
NFR1	Performance	The system shall detect and report an intrusion event within 5 seconds of a sensor trigger.
NFR2	Reliability	The system shall maintain 99.9% uptime and continue local operation during internet outages.
NFR3	Security	All data transmitted between devices and the cloud shall be encrypted using TLS/SSL protocols.
NFR4	Scalability	The system shall support the addition of new sensors and devices without requiring system redesign.
NFR5	Usability	The mobile application shall be intuitive and operable by non-technical users.
NFR6	Power Efficiency	Battery-powered sensors shall operate for a minimum of 12 months on a single charge.
NFR7	Privacy	Video footage and personal data shall be stored securely and accessible only to authorized users.
NFR8	Maintainability	The system shall support over-the-air (OTA) firmware updates for all connected devices.
NFR9	Interoperability	The system shall be compatible with common smart home platforms (e.g., Google Home, Amazon Alexa).
NFR10	Response Time	The mobile application shall load a live camera feed within 3 seconds under normal network conditions.

Step 2: Process Specification

This section details the core processes performed by SHIDS, followed by flowcharts that illustrate the logic and flow of the two most critical processes.

2.1 System Processes

2.1.1 Process 1: Intrusion Detection

The system continuously polls all door/window contact sensors and motion sensors. When a sensor is triggered, the system checks whether the alarm is currently armed. If armed, it initiates the intrusion response pipeline. If disarmed, the event is silently logged for record-keeping.

2.1.2 Process 2: Alert & Notification

Upon confirming an intrusion, the system simultaneously triggers the local siren, captures a snapshot or video clip from the nearest camera, and pushes a real-time notification to the homeowner's mobile application. If the homeowner does not respond within a configurable timeout (e.g., 60 seconds), the system automatically escalates by notifying emergency contacts or authorities.

2.1.3 Process 3: Remote Arm/Disarm

The homeowner can arm or disarm the system at any time through the mobile application. The command is sent to the IoT gateway via the cloud, which relays it to all connected sensors and actuators. The system confirms the state change and notifies the user accordingly.

2.1.4 Process 4: Live Monitoring

The homeowner can request a live video stream from any connected camera at any time. The camera feed is routed through the gateway to the cloud and delivered to the mobile application in real time, enabling continuous remote surveillance.

2.1.5 Process 5: Event Logging

Every system event — including sensor triggers, arm/disarm actions, alerts sent, and camera snapshots — is timestamped and stored in the cloud database. The homeowner can review the full event history through the mobile application at any time.

2.1.6 Process 6: Smart Lock Control

The homeowner can remotely lock or unlock smart door locks through the mobile application. In the event of a confirmed intrusion, the system can automatically lock all doors as a containment measure to restrict unauthorized movement within the premises.

2.1.7 Process 7: OTA Firmware Update

When a firmware update is available, the IoT gateway notifies all connected devices and coordinates the update process sequentially, ensuring no devices are left simultaneously offline during the update cycle.

2.2 Process Flowcharts

2.2.1 Flowchart 1: Intrusion Detection & Alert Process

Figure 1 illustrates the end-to-end logic of the intrusion detection and alert notification process, which represents the core operational flow of SHIDS.

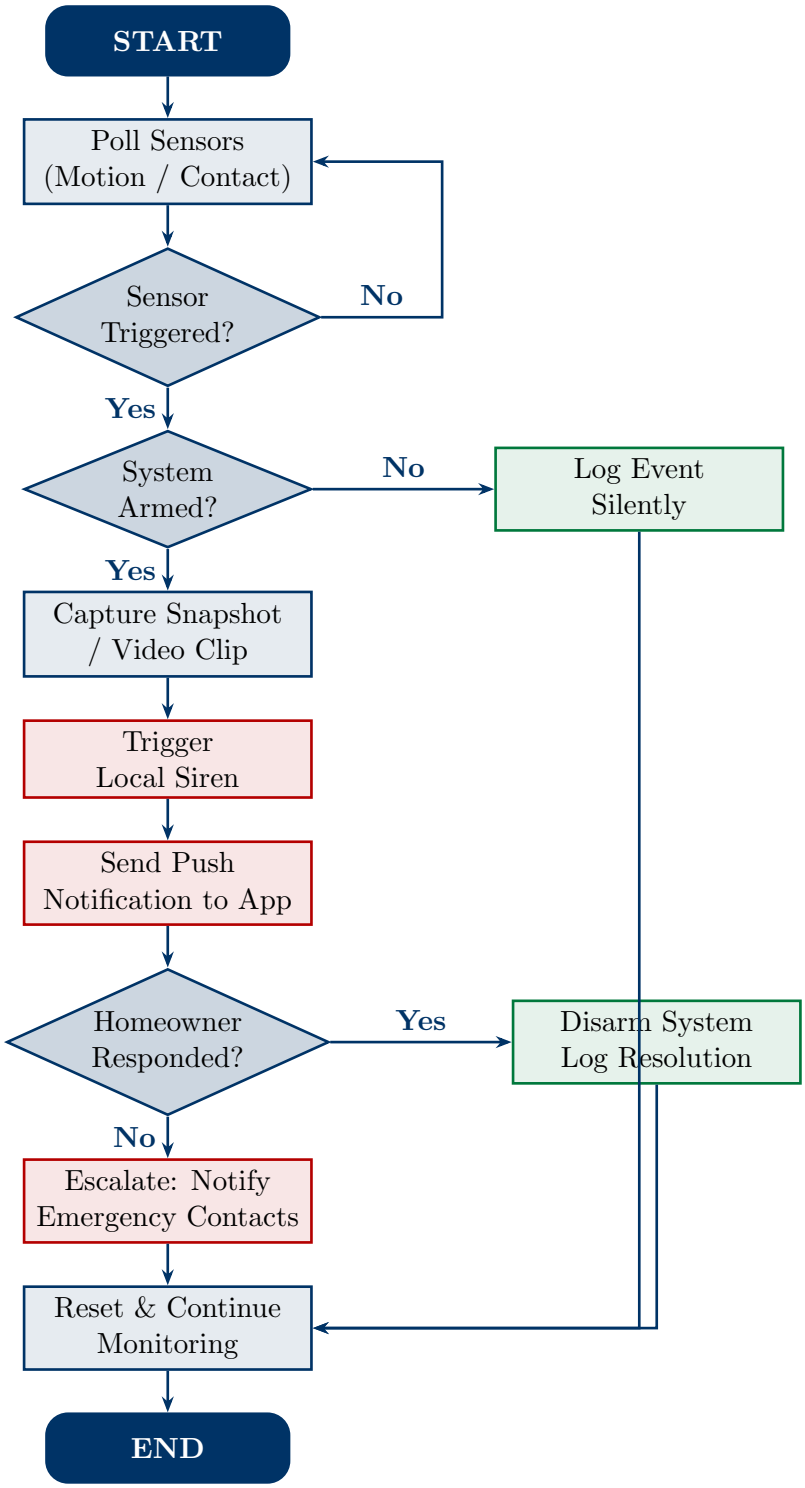

Figure 1: Flowchart of the Intrusion Detection and Alert Process

2.2.2 Flowchart 2: Overall System Process Flow

Figure 2 provides a high-level overview of all seven processes within SHIDS, showing how each process is triggered and how the devices interact with the central gateway and cloud platform.

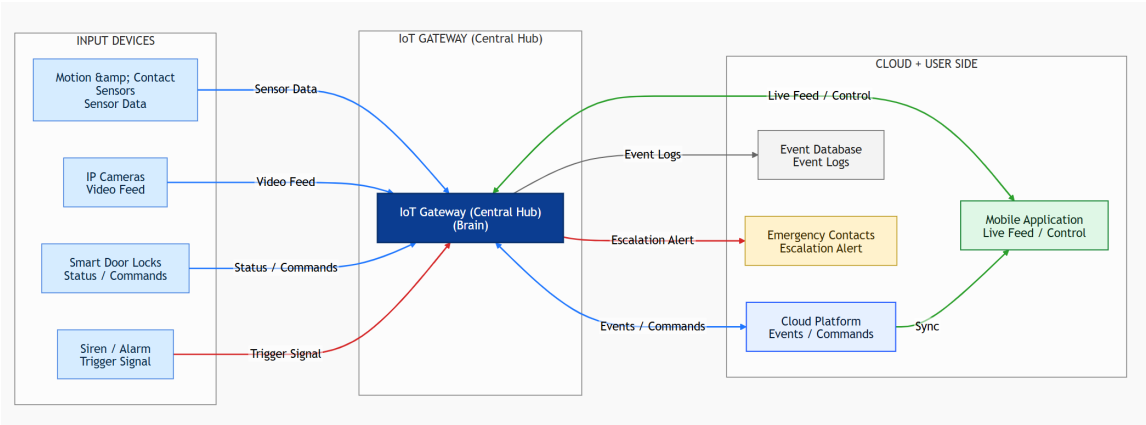

Figure 2: High-Level System Process Flow of SHIDS

Step 3: Domain Model Specification

The domain model identifies the key real-world entities involved in SHIDS and defines the relationships between them. Each entity represents a distinct component or actor within the system. Figure 4 presents the UML-style domain model diagram.

3.1 Key Entities

- **Homeowner** — The primary user of the system. Interacts with SHIDS via the mobile application to arm/disarm the system, view live feeds, and receive security alerts.
- **Mobile Application** — The user-facing interface through which the homeowner sends commands and receives notifications. Communicates bidirectionally with the cloud platform.
- **IoT Gateway** — The central processing hub. Collects sensor data, executes local logic, controls actuators, and bridges all devices with the cloud platform.
- **Motion Sensor** — Detects movement within a monitored zone and reports trigger events to the gateway in real time.
- **Contact Sensor** — Monitors the open/closed state of doors and windows, reporting any state change to the gateway immediately.
- **IP Camera** — Captures images and video upon trigger and streams live footage through the gateway to the cloud platform.
- **Smart Door Lock** — Receives lock/unlock commands from the gateway and reports its current state back to the system.
- **Siren / Alarm** — An actuator activated by the gateway to produce an audible alert upon confirmed intrusion.
- **Cloud Platform** — Receives and processes all data from the gateway, stores events, and relays commands and notifications to the mobile application.
- **Event Database** — A persistent storage component that logs all system events including sensor triggers, snapshots, arm/disarm records, and alert histories.
- **Emergency Contact** — A designated person or authority automatically notified by the cloud platform when the homeowner fails to respond to an intrusion alert.

3.2 Entity Relationships

- A **Homeowner** uses exactly one **Mobile Application**.
- A **Mobile Application** communicates bidirectionally with one **Cloud Platform**.
- A **Cloud Platform** communicates bidirectionally with one **IoT Gateway**.

- An **IoT Gateway** manages one or more **Motion Sensors**, **Contact Sensors**, **IP Cameras**, and **Smart Door Locks**, and controls exactly one **Siren**.
- A **Cloud Platform** writes all events to one **Event Database**.
- A **Cloud Platform** notifies one or more **Emergency Contacts** upon escalation.

3.3 UML Domain Model Diagram

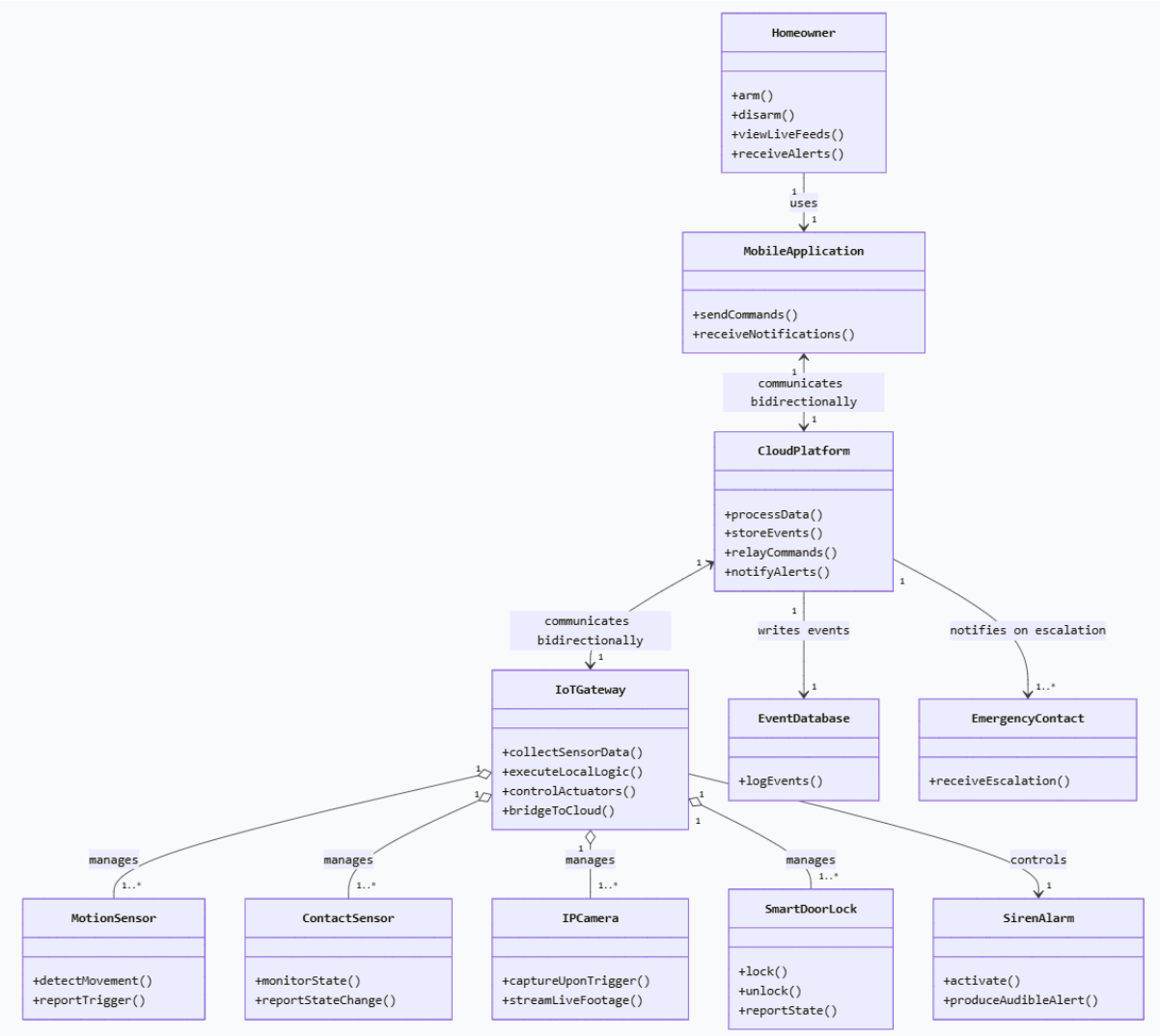

Figure 3: UML Domain Model Diagram of SHIDS

Step 4: Information Model Specification

This section specifies the data that will be used, generated, and managed by SHIDS, detailing the structure of each data entity and the flow of information across system components.

4.1 Data Entities & Structure

The following tables describe the six key data entities managed by SHIDS.

4.1.1 Entity 1: Sensor Event

A **Sensor Event** is generated whenever a motion or contact sensor is triggered. It captures the identity, location, and timing of the physical event detected by the sensor.

Table 3: Sensor Event Data Entity

Field	Data Type	Description
event_id	String	Unique identifier for the event.
sensor_id	String	Identifier of the sensor that triggered the event.
sensor_type	Enum	Type of sensor: Motion or Contact.
location	String	Physical location of the sensor (e.g., Front Door).
timestamp	DateTime	Date and time the trigger occurred.
status	Enum	Current sensor state: Triggered or Idle.

4.1.2 Entity 2: Intrusion Alert

An **Intrusion Alert** is generated by the cloud platform when a sensor event is confirmed as a potential intrusion. It tracks the alert lifecycle from generation to resolution.

Table 4: Intrusion Alert Data Entity

Field	Data Type	Description
alert_id	String	Unique identifier for the alert.
event_id	String	Reference to the triggering sensor event.
alert_level	Enum	Severity level: Low, Medium, or High.
timestamp	DateTime	Time the alert was generated.
response_status	Enum	Alert state: Pending, Acknowledged, or Escalated.
camera_snapshot	Binary	Image captured at the time of the alert.

4.1.3 Entity 3: Camera Feed

A **Camera Feed** record is created whenever a live stream or recording session is initiated by the system or the homeowner.

Table 5: Camera Feed Data Entity

Field	Data Type	Description
feed_id	String	Unique identifier for the feed session.
camera_id	String	Identifier of the camera.
location	String	Physical location of the camera.
stream_url	String	URL of the live video stream.
recording_path	String	Cloud storage path of the saved recording.
timestamp	DateTime	Feed session start time.

4.1.4 Entity 4: User Command

A **User Command** is issued by the homeowner through the mobile application and delivered to the relevant device via the cloud and gateway.

Table 6: User Command Data Entity

Field	Data Type	Description
command_id	String	Unique identifier for the command.
user_id	String	Identifier of the homeowner issuing the command.
command_type	Enum	Type of command: Arm, Disarm, Lock, Unlock, or Stream.
target_device	String	Identifier of the target device.
timestamp	DateTime	Time the command was issued.
status	Enum	Delivery state: Sent, Delivered, Executed, or Failed.

4.1.5 Entity 5: Device Status

A **Device Status** record represents the current operational state of any connected device within SHIDS, updated periodically by each device.

4.1.6 Entity 6: Event Log

An **Event Log** entry is written to the cloud database for every significant system event, providing a complete and auditable history of system activity.

Table 7: Device Status Data Entity

Field	Data Type	Description
device_id	String	Unique identifier for the device.
device_type	Enum	Device category: Sensor, Camera, Lock, Siren, or Gateway.
status	Enum	Operational state: Online, Offline, or Fault.
battery_level	Integer	Remaining battery percentage (0–100).
firmware_version	String	Currently installed firmware version.
last_updated	DateTime	Timestamp of the most recent status update.

Table 8: Event Log Data Entity

Field	Data Type	Description
log_id	String	Unique identifier for the log entry.
event_type	String	Category of the event: Alert, Command, or Status.
source	String	Component that generated the event.
description	String	Human-readable description of the event.
timestamp	DateTime	Time the event was recorded.

4.2 Information Flow

Information flows through SHIDS in a structured pipeline across six stages, as described below and illustrated in Figure **??**.

1. **Sensing** — Motion and contact sensors detect physical events and transmit raw sensor event data to the IoT Gateway.
2. **Local Processing** — The IoT Gateway processes incoming sensor data, evaluates the system arm state, and forwards structured events to the Cloud Platform.
3. **Cloud Processing & Storage** — The Cloud Platform receives events, stores them in the Event Database, generates intrusion alerts where applicable, and dispatches push notifications to the Mobile Application.
4. **User Interaction** — The Mobile Application presents alerts and live feeds to the Homeowner, who can issue commands such as arm, disarm, lock, or unlock.
5. **Command Execution** — User commands travel from the Mobile Application through the Cloud Platform to the IoT Gateway, which relays them to the appropriate actuator (smart lock or siren).
6. **Logging** — All events at every stage are timestamped and written to the Event Database for audit, review, and historical analysis.

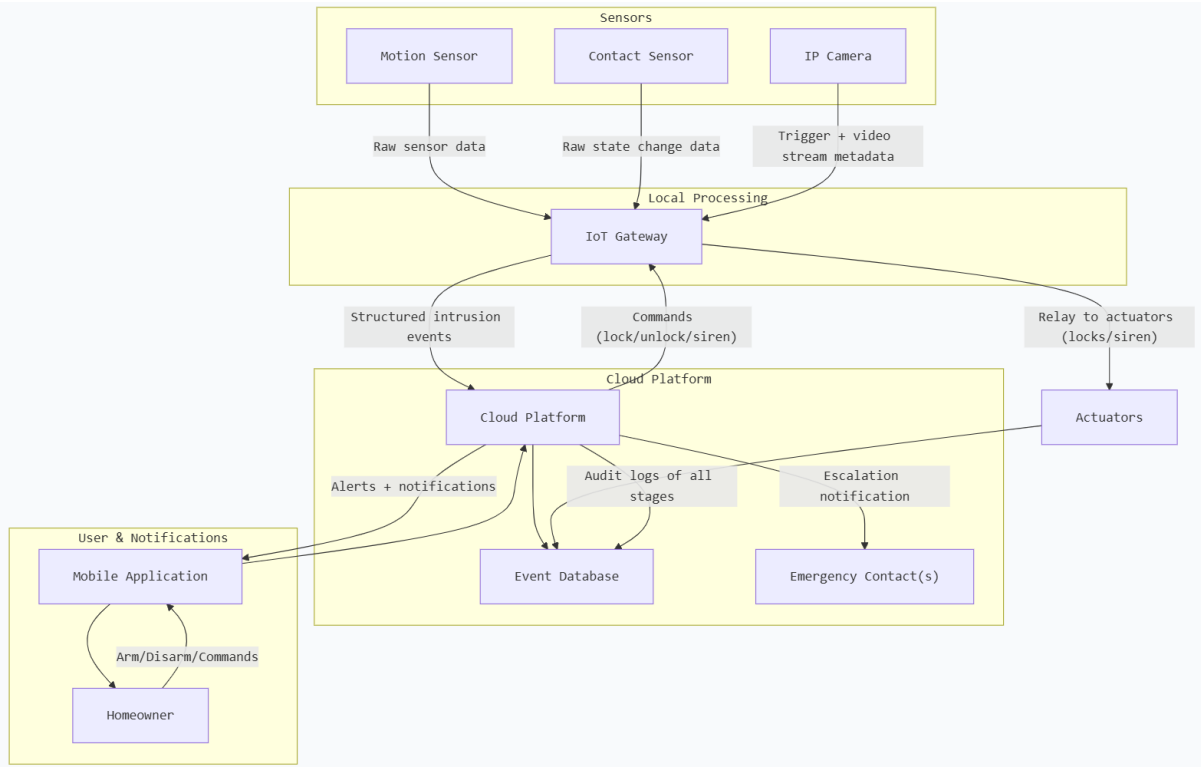

Figure 4: Information Flow Diagram of SHIDS

Step 5: Service Specification

This section defines the set of services provided by the Smart Home Intrusion Detection System (SHIDS). These services represent the functional building blocks of the system and enable interaction between sensors, actuators, the IoT gateway, cloud infrastructure, and the mobile application.

Each service exposes well-defined interfaces and operations to ensure modularity, scalability, and efficient communication across system components.

5.1 Overview of Services

SHIDS provides a set of distributed services that operate collaboratively to achieve real-time intrusion detection, user notification, device control, and system monitoring. These services are implemented across both edge (IoT Gateway) and cloud layers to ensure low latency and high availability.

5.2 Core Services

The system provides the following five core services:

- Intrusion Detection Service
- Notification Service
- Video Monitoring Service

- Device Control Service
- Data Logging Service

5.3 Detailed Service Specifications

5.3.1 Service 1: Intrusion Detection Service

Description: Continuously monitors sensor data from motion and contact sensors to detect unauthorized activity. When the system is armed, it initiates the intrusion response process.

Service Interface:

- Input: Sensor Event Data
- Output: Intrusion Alert, Trigger Signal

Operations:

- detectIntrusion(event)
- checkSystemState()
- triggerAlarm()

5.3.2 Service 2: Notification Service

Description: Responsible for delivering real-time alerts and notifications to the homeowner via the mobile application.

Service Interface:

- Input: Intrusion Alert
- Output: Push Notification

Operations:

- sendAlert(alert)
- notifyUser(userID)
- escalateAlert(alert)

5.3.3 Service 3: Video Monitoring Service

Description: Handles real-time video streaming and recording from IP cameras, supporting both live monitoring and event-triggered capture.

Service Interface:

- Input: Stream Request / Sensor Trigger
- Output: Live Video Feed / Recorded Video

Operations:

- startStream(cameraID)
- stopStream(cameraID)
- captureSnapshot(cameraID)

5.3.4 Service 4: Device Control Service

Description: Executes user commands such as arming/disarming the system and controlling smart locks.

Service Interface:

- Input: User Command
- Output: Device Status Update

Operations:

- armSystem()
- disarmSystem()
- lockDoor(deviceID)
- unlockDoor(deviceID)

5.3.5 Service 5: Data Logging Service

Description: Records all system events, alerts, and user actions into the cloud database for auditing and analysis.

Service Interface:

- Input: System Events
- Output: Stored Logs

Operations:

- logEvent(event)
- retrieveLogs(userID)
- archiveLogs()

Step 6: IoT Level Specification

6.1 Overview of IoT Architecture Levels

The Smart Home Intrusion Detection System (SHIDS) conforms to the **six-level IoT design reference model**, which organises the system into hierarchical, interoperable layers. Each level has a clearly defined responsibility, a specific set of technologies, and well-specified interfaces to adjacent levels. Table 9 summarises the six levels as applied to SHIDS, and Figure 5 illustrates their interactions.

Table 9: IoT Level Specification for SHIDS

Level	Name	Role in SHIDS
1	Physical Devices & Controllers	Motion sensors (PIR), door/window contact sensors, IP cameras, smart door locks, and the siren actuator. These devices sense the physical environment and execute actuation commands.
2	Connectivity	Wi-Fi (IEEE 802.11n/ac) links all sensors and actuators to the IoT gateway. MQTT over TCP/IP carries lightweight sensor telemetry; RTSP streams live camera video to the gateway.
3	Edge Computing (Fog)	The ESP32-based IoT gateway performs local event filtering, threshold-based intrusion logic, and alarm triggering without cloud dependency. This ensures sub-5-second response even during internet outages.
4	Data Accumulation	Validated intrusion events, snapshots, and system state changes are buffered at the gateway and forwarded via HTTPS/MQTT to the cloud platform for persistent storage in a time-series event database.
5	Data Abstraction & Processing	The cloud back-end aggregates, indexes, and analyses stored events. It applies rule-based logic to escalate unacknowledged alerts to emergency contacts and exposes a RESTful API consumed by the mobile application.
6	Application	The mobile and web applications provide the homeowner with real-time dashboards, live camera feeds (via WebRTC), historical event logs, remote arm/disarm controls, and smart lock management.

6.2 Technology Stack per Level

Table 10: Technology Mapping Across IoT Levels

Level	Layer	Technologies & Protocols
1	Sensing & Actuation	PIR sensors, magnetic contact sensors, CMOS IP cameras, ESP32/Arduino microcontrollers, siren module
2	Connectivity	Wi-Fi 802.11n/ac, MQTT (port 1883/8883), RTSP, ZigBee (optional battery sensors)
3	Edge Computing	ESP32 firmware (C/C++), local MQTT broker (Mosquitto), threshold-based decision engine, local SQLite buffer
4	Data Accumulation	HTTPS REST (TLS 1.3), MQTT over WSS, AWS IoT Core / Azure IoT Hub ingestion pipeline
5	Processing	Cloud microservices (Node.js), PostgreSQL time-series DB, rule engine for escalation, RESTful API (JSON)
6	Application	React Native mobile app, WebRTC live streaming, Firebase push notifications (FCM/APNs)

6.3 IoT Level Interaction Diagram

Figure 5 illustrates the vertical data flow across all six IoT levels of SHIDS, from raw sensor data at the device layer to user-facing services at the application layer, including feedback control paths.

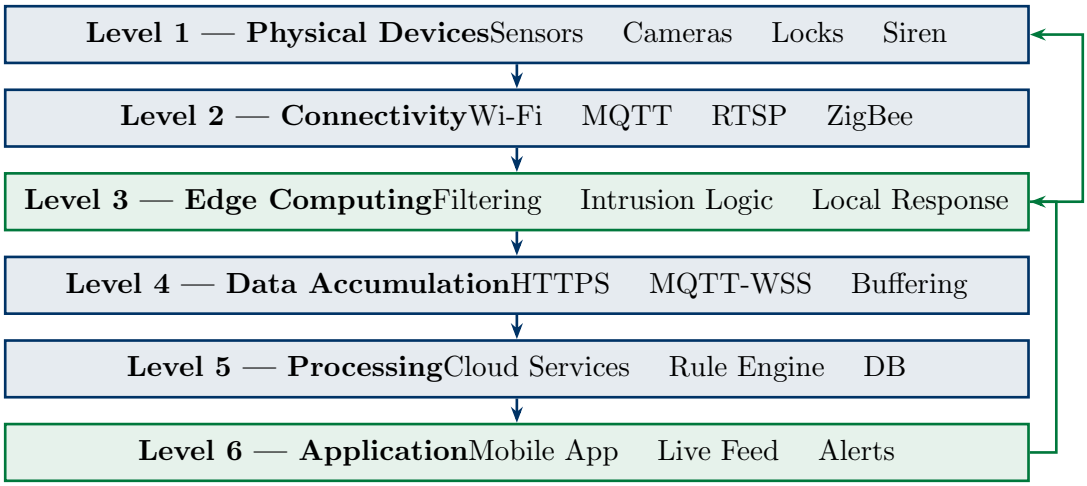

Figure 5: Six-Level IoT Architecture of SHIDS showing data flow and control feedback

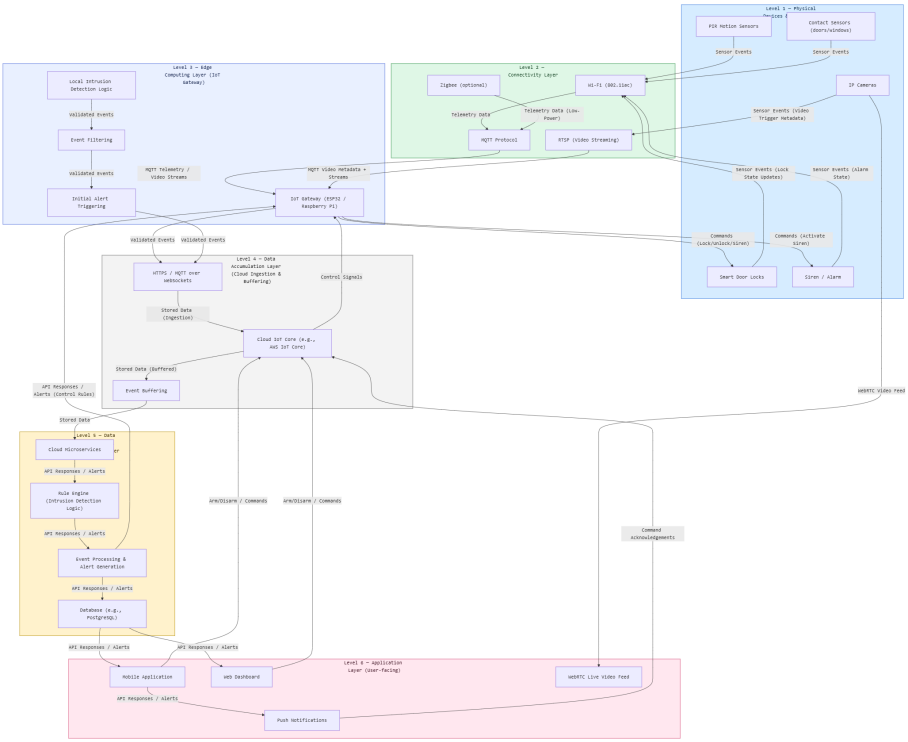

Figure 6: Enhanced IoT Interaction Diagram

6.4 Justification of Design Choices

- **Edge computing at Level 3** is critical for SHIDS because intrusion response must occur within 5 seconds regardless of cloud availability. Placing decision logic at the gateway eliminates round-trip latency and ensures operation during internet outages (NFR2).
- **MQTT** is chosen at Level 2 over HTTP polling because it operates on a publish/subscribe model, reducing bandwidth by up to 93% compared to HTTP for small payloads, and supports QoS levels 1 and 2 to guarantee message delivery.
- **ZigBee** is offered as an optional protocol for battery-powered contact sensors because its mesh topology and low duty cycle allow battery life exceeding 12 months (NFR6), while Wi-Fi sensors can draw power from mains-connected positions.
- **WebRTC** at Level 6 provides peer-to-peer live video streaming with sub-3-second latency directly to the mobile application, satisfying NFR10, without requiring proprietary media servers.

Step 7: Communication Model Specification

7.1 Overview

The communication model of the Smart Home Intrusion Detection System (SHIDS) defines how data is transmitted between system components, including sensors, the IoT gateway, cloud infrastructure, and the mobile application. The system follows a hybrid communication approach combining lightweight messaging protocols, real-time streaming technologies, and secure web-based APIs to ensure efficient, reliable, and low-latency communication.

The communication model is designed to support real-time intrusion detection, instant user notification, and remote control of devices while maintaining high security and scalability.

7.2 Communication Components

The SHIDS communication architecture consists of the following main components:

- **Physical Devices** — Motion sensors, contact sensors, IP cameras, and smart door locks that generate and receive data.
- **IoT Gateway** — Acts as the central communication hub that aggregates device data, performs local processing, and communicates with the cloud.
- **Cloud Platform** — Handles data storage, processing, notification services, and API endpoints.
- **Mobile Application** — Provides the user interface for monitoring, receiving alerts, and issuing commands.

7.3 Communication Protocols

SHIDS employs multiple communication protocols, each selected based on the requirements of specific data types and system constraints.

7.4 Communication Patterns

SHIDS follows a combination of communication patterns to support different system operations:

- **Publish/Subscribe (MQTT)** — Sensors publish data to topics, and the IoT gateway subscribes to receive updates in real time.
- **Request/Response (HTTP/HTTPS)** — The mobile application sends commands and requests to the cloud via RESTful APIs.
- **Event-Driven Communication** — Intrusion events trigger automated workflows, including alerts and actuation.
- **Streaming Communication** — Cameras stream live video using RTSP and WebRTC protocols.

Table 11: Communication Protocols Used in SHIDS

Protocol	Usage	Justification
MQTT	Sensor data transmission	Lightweight, low bandwidth, publish/subscribe model suitable for IoT devices
HTTP/HTTPS	API communication	Reliable request/response model with secure data transfer using TLS
RTSP	Video streaming	Enables real-time transmission of camera feeds with minimal delay
WebRTC	Live video delivery	Provides low-latency peer-to-peer streaming to mobile applications
TLS/SSL	Security layer	Ensures encryption and secure communication across all channels

7.5 Message Flow Description

The communication flow in SHIDS is bidirectional and can be described through typical system scenarios:

7.5.1 Scenario 1: Intrusion Detection and Alert

1. A motion or contact sensor detects an event and publishes data via MQTT.
2. The IoT gateway receives the message and processes it locally.
3. If an intrusion is detected, the gateway forwards the event to the cloud.
4. The cloud platform generates an intrusion alert and sends a push notification to the mobile application.
5. The homeowner receives the alert and may respond through the application.

7.5.2 Scenario 2: Remote Command Execution

1. The homeowner sends a command (e.g., arm/disarm, lock/unlock) via the mobile application.
2. The command is transmitted to the cloud using HTTPS.
3. The cloud forwards the command to the IoT gateway.
4. The gateway relays the command to the target device (e.g., smart lock or siren).
5. The device executes the command and updates its status.

7.6 Communication Model Diagram

Figure 7 illustrates the communication architecture of SHIDS, highlighting data flows between devices, the gateway, the cloud, and the mobile application.

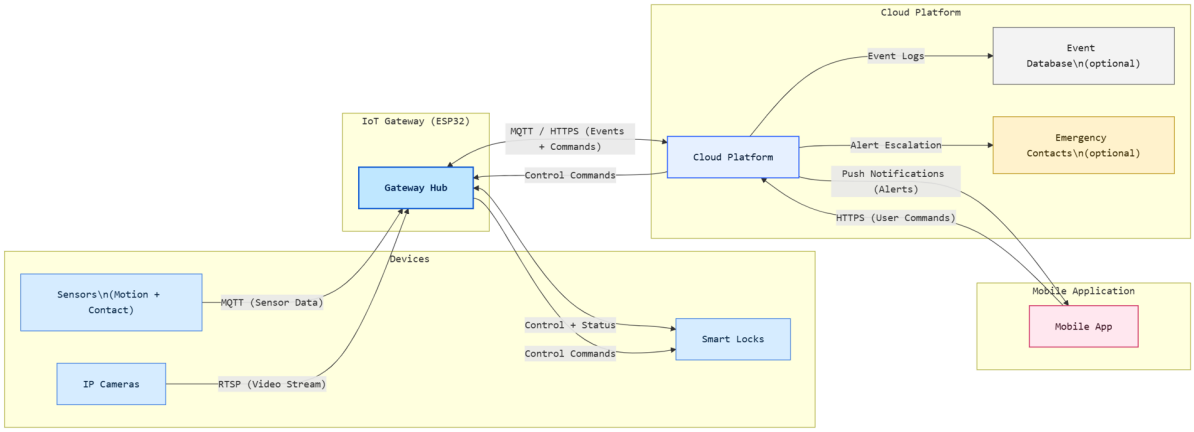

Figure 7: Communication Model of SHIDS

7.7 Security Considerations

Security is a critical aspect of the SHIDS communication model. The system implements multiple security mechanisms:

- **Encryption** — All communications are secured using TLS/SSL to prevent eavesdropping and data interception.
- **Authentication** — Devices and users are authenticated before accessing system resources.
- **Access Control** — Role-based access ensures only authorized users can issue commands or view data.
- **Data Integrity** — Messages are protected against tampering during transmission.

7.8 Justification of Design Choices

The chosen communication model ensures:

- **Low Latency** — MQTT and edge processing enable rapid response to intrusion events.
- **Scalability** — The publish/subscribe model allows seamless addition of new devices.
- **Reliability** — Cloud-based processing ensures consistent system availability.
- **Security** — End-to-end encryption and authentication protect user data and system integrity.

Step 8: Deployment & Operational View

8.1 Overview

The deployment and operational view of the Smart Home Intrusion Detection System (SHIDS) describes how system components are physically distributed and how they operate in a real-world environment. The system is deployed across multiple layers, including the home environment, edge computing layer, cloud infrastructure, and user interface layer.

This deployment ensures real-time responsiveness, high availability, and secure communication between all components, enabling continuous monitoring and control of the smart home system.

8.2 Deployment Layers

The SHIDS architecture is deployed across four primary layers:

- **Home Environment Layer** — Contains all physical devices installed within the residential premises, including motion sensors, contact sensors, IP cameras, smart door locks, and the siren actuator.
- **Edge Layer (IoT Gateway)** — The ESP32-based IoT gateway is deployed within the home network and acts as the central controller. It aggregates sensor data, performs local processing, and ensures immediate response to intrusion events.
- **Cloud Infrastructure Layer** — Hosted on platforms such as AWS or Azure, this layer manages data storage, processing, analytics, and notification services.
- **Application Layer** — The mobile application installed on the homeowner's device provides real-time monitoring, alerts, and remote control capabilities.

8.3 Physical Deployment Components

8.3.1 Home Environment

The home environment consists of interconnected IoT devices deployed at strategic locations:

- Motion sensors and contact sensors installed at entry points.
- IP cameras positioned to monitor critical areas.
- Smart door locks installed on main access doors.
- A siren/alarm unit for audible alerts.

All devices communicate wirelessly with the IoT gateway using Wi-Fi or ZigBee protocols.

8.3.2 IoT Gateway

The IoT gateway (ESP32) is deployed within the local network and serves as the edge computing unit. It performs the following functions:

- Aggregates data from all connected devices.
- Executes local intrusion detection logic.
- Triggers immediate responses such as activating the siren.
- Communicates with the cloud platform for further processing.

8.3.3 Cloud Infrastructure

The cloud layer is responsible for centralized system management and includes:

- Cloud server hosting backend services and APIs.
- Event database for storing logs, alerts, and historical data.
- Notification service for delivering real-time alerts.

8.3.4 User Application

The mobile application operates on the homeowner's smartphone and provides:

- Real-time notifications of intrusion events.
- Live video streaming from cameras.
- Remote control of system functions (arm/disarm, lock/unlock).
- Access to historical logs and system status.

8.4 Network Architecture

The system operates over a hybrid network architecture:

- **Local Network** — Devices communicate with the IoT gateway via Wi-Fi or ZigBee.
- **Internet Connectivity** — The gateway connects to the cloud via the internet using secure protocols.
- **Secure Communication** — All external communications are protected using TLS/SSL encryption.

8.5 Operational Behavior

The SHIDS operates continuously with the following behavior:

- Devices remain active and continuously monitor the environment.
- The IoT gateway processes incoming data in real time.
- The cloud platform stores data and manages system-wide logic.
- The mobile application interacts with the system on demand.

This design ensures that intrusion detection and response can occur even during temporary internet outages due to local processing at the gateway.

8.6 Deployment Diagram

Figure 8 illustrates the physical deployment architecture of SHIDS, showing the interaction between home devices, the IoT gateway, cloud infrastructure, and the mobile application.

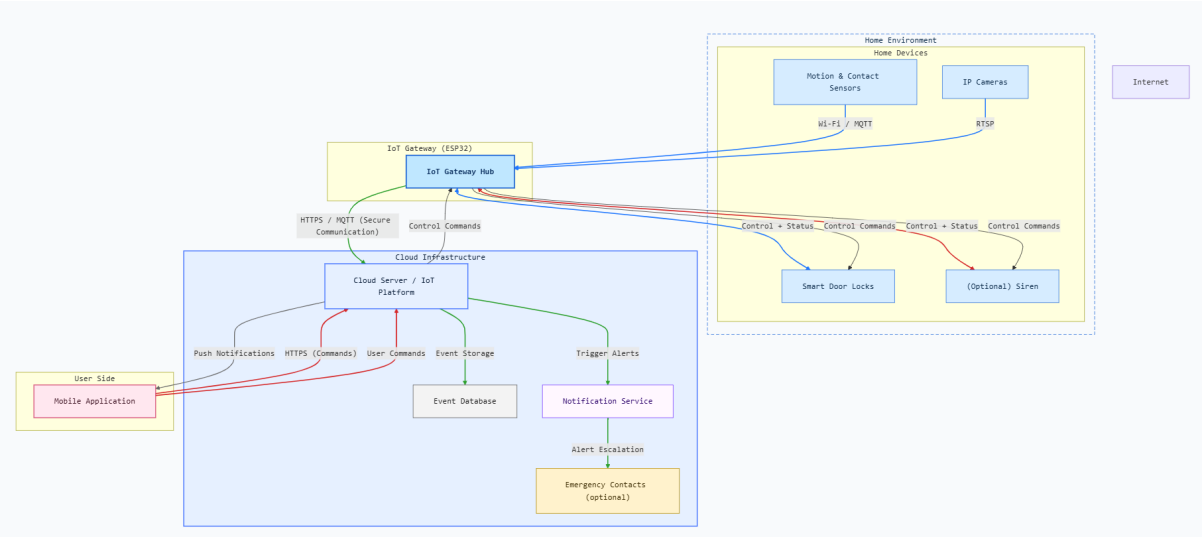

Figure 8: Deployment Architecture of SHIDS

8.7 Justification of Deployment Design

The deployment strategy of SHIDS is designed to achieve optimal system performance:

- **Low Latency** — Edge processing at the gateway enables immediate response to intrusion events.
- **Scalability** — Cloud infrastructure allows seamless addition of new devices and users.
- **Reliability** — Local operation ensures system functionality even during internet outages.
- **Security** — End-to-end encryption protects all communication channels and user data.

Step 9: Business Model & Feasibility Analysis

9.1 Overview

The Smart Home Intrusion Detection System (SHIDS) is designed as a cost-effective, scalable, and intelligent home security solution targeting residential users. This section evaluates the economic viability, deployment feasibility, and potential business model of the system, ensuring that it is not only technically sound but also practical for real-world adoption.

9.2 Target Market

SHIDS is primarily targeted at:

- Homeowners seeking affordable and reliable security systems.
- Apartment residents requiring remote monitoring capabilities.
- Property owners managing multiple residential units.

The system is particularly attractive to users who want smart security features without the high costs of traditional security systems.

9.3 Cost Analysis

9.3.1 Hardware Cost Estimation

Table 12 provides an estimated cost breakdown for a typical SHIDS deployment in a single household.

Table 12: Estimated Hardware Cost per Household

Component	Quantity	Estimated Cost (USD)
Motion / Contact Sensors	4–6	20–40
IP Camera	1–2	30–80
Smart Door Lock	1	50–100
ESP32 IoT Gateway	1	10–15
Siren / Alarm	1	10–15
Total Estimated Cost		**120–250 USD**

9.3.2 Cloud and Operational Costs

The cloud infrastructure introduces recurring operational costs, including:

- Cloud hosting (AWS / Azure)
- Database storage and backups

- Notification services (push notifications, messaging)

These costs are relatively low for small-scale deployments and increase gradually with the number of users, making the system highly scalable.

9.3.3 Development and Maintenance Costs

Initial development costs include:

- Mobile application development
- Backend cloud services implementation
- Firmware development for IoT devices

Ongoing maintenance includes software updates, security patches, and cloud service management.

9.4 Revenue Model

The SHIDS system can adopt a hybrid revenue model combining hardware sales and subscription-based services:

- **One-Time Hardware Purchase** — Users purchase the physical devices (sensors, cameras, gateway, etc.).
- **Subscription Services** — Optional monthly or yearly subscription for advanced features such as cloud video storage, extended event history, and emergency alert services.
- **Freemium Model** — Basic features (real-time alerts and monitoring) are provided for free, while premium features require a paid subscription.

This model ensures accessibility for basic users while generating recurring revenue from advanced services.

9.5 Feasibility Analysis

9.5.1 Technical Feasibility

The system is highly feasible from a technical perspective, as it relies on well-established technologies such as IoT sensors, cloud computing, mobile applications, and wireless communication protocols.

9.5.2 Economic Feasibility

The relatively low hardware cost and scalable cloud infrastructure make SHIDS economically viable for both users and service providers. The subscription model provides sustainable long-term revenue.

9.5.3 Operational Feasibility

The system is designed to be user-friendly and requires minimal technical knowledge for operation. Installation and maintenance are straightforward, making it suitable for widespread adoption.

9.6 Advantages

- Real-time intrusion detection and alerts.
- Remote monitoring and control via mobile application.
- Scalable and modular system design.
- Cost-effective compared to traditional security systems.

9.7 Limitations

- Dependence on internet connectivity for cloud-based features.
- Potential privacy concerns related to video monitoring.
- Initial setup cost may be a barrier for some users.

Step 10: Deployment & Maintenance Specification

10.1 Deployment Architecture

The Smart Home Intrusion Detection System (SHIDS) follows a hybrid deployment model that combines edge devices, cloud infrastructure, and user-facing applications.

- **Edge Layer Deployment** The IoT Gateway (ESP32) is deployed inside the home and acts as the central hub. All sensors (motion, contact), cameras, and actuators (siren, smart locks) are connected to the gateway via Wi-Fi or ZigBee.
- **Cloud Layer Deployment** The backend services are deployed on a cloud platform (e.g., AWS IoT Core or Azure IoT Hub). These services handle data ingestion, event processing, alert generation, and storage.
- **Application Layer Deployment** The mobile application is deployed on Android and iOS using React Native. Users install it and connect it to their home system.

10.2 User Interface Screens

This section presents sample user interface screens of the SHIDS mobile application, demonstrating how users interact with the system.

Figure 9: Main Dashboard Screen (System Status Overview)

Figure 10: Live Camera Feed Interface

Figure 11: Intrusion Alert Notification Screen

Figure 12: Device Control (Arm/Disarm & Smart Lock)

10.3 Network Configuration

- Devices connect to the gateway via local Wi-Fi.
- The gateway communicates with the cloud using:

  - MQTT (sensor data)
  - HTTPS (API requests)
- TLS encryption ensures secure communication.
- Ports used:
  - MQTT: 1883 / 8883
  - HTTPS: 443

10.4 Installation Procedure

1. Install and power the IoT Gateway.
2. Connect it to the home Wi-Fi network.
3. Pair sensors and devices with the gateway.
4. Install the mobile application.
5. Register user account and link the system.
6. Configure zones, alerts, and preferences.

10.5 Maintenance Strategy

- OTA firmware updates for all devices.
- Continuous health monitoring (battery, connectivity).
- Automatic fault detection and recovery.
- Event logging for auditing and debugging.

10.6 Scalability Considerations

- Easy addition of new sensors/devices.
- Cloud auto-scaling under load.
- Modular microservices architecture.

10.7 Security Measures

- TLS encryption for all communications.
- Secure authentication (tokens).
- Role-based access control.
- Secure credential storage.

Conclusion

This report presented the complete design of the Smart Home Intrusion Detection System (SHIDS), covering all major aspects of an IoT-based solution, including system requirements, process specification, domain and information models, service architecture, IoT layered design, and deployment strategy.

The proposed system demonstrates how IoT technologies can be effectively integrated to provide real-time monitoring, intelligent intrusion detection, and automated response mechanisms. By combining edge computing at the gateway with cloud-based processing and a user-friendly mobile application, SHIDS achieves both low-latency operation and high scalability.

Overall, the system provides a reliable, secure, and cost-effective solution for modern smart home security, while remaining flexible for future expansion and integration with additional smart devices and services.